

Module 01: Grids, Galaxies, and Ray-Tracing

Theory Companion — Learning to Autolens

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Abstract

This document provides the mathematical background for Module 01 of *Learning to Autolens*. We derive the gravitational lens equation, define the key lensing quantities (convergence, shear, magnification), and describe the mass and light profile models used in PyAutoLens. All results are self-contained; companion references are Congdon & Keeton (2018, hereafter C&K), Narayan & Bartelmann (1997, hereafter N&B), and Schneider, Ehlers & Falco (1992, hereafter S92).

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1 The Gravitational Lens Equation

The fundamental equation of gravitational lensing maps angular positions in the **image plane** θ to the **source plane** β via the scaled deflection angle α :

Key Result

$$\beta = \theta - \alpha(\theta) \quad (1)$$

where θ and β are two-dimensional angular positions (in arc-seconds), and $\alpha(\theta)$ is the scaled deflection angle at position θ (C&K eq. 4.1; N&B eq. 3).

The deflection angle is related to the physical deflection $\hat{\alpha}$ by the distance ratio:

$$\alpha = \frac{D_{\text{ds}}}{D_{\text{s}}} \hat{\alpha} \quad (2)$$

where D_{d} , D_{s} , and D_{ds} are the angular diameter distances to the lens, to the source, and between the lens and source, respectively.

1.1 Critical Surface Density

The strength of lensing is governed by the **critical surface mass density**:

$$\Sigma_{\text{cr}} = \frac{c^2}{4\pi G} \frac{D_{\text{s}}}{D_{\text{d}} D_{\text{ds}}} \quad (3)$$

(C&K eq. 4.5; N&B eq. 10). This sets the scale: a lens with projected surface density $\Sigma \geq \Sigma_{\text{cr}}$ produces multiple images (strong lensing).

1.2 Einstein Radius

The **Einstein radius** is the characteristic angular scale of strong lensing:

$$\theta_{\text{E}} = \sqrt{\frac{4GM(\theta_{\text{E}})}{c^2} \frac{D_{\text{ds}}}{D_{\text{d}} D_{\text{s}}}} \quad (4)$$

where $M(\theta_{\text{E}})$ is the projected mass enclosed within θ_{E} (C&K eq. 4.9). For a typical galaxy-scale lens, $\theta_{\text{E}} \sim 0.5''\text{--}3''$.

2 Convergence, Shear, and Magnification

2.1 Convergence

The **convergence** is the dimensionless projected surface mass density:

$$\kappa(\theta) = \frac{\Sigma(\theta)}{\Sigma_{\text{cr}}} \quad (5)$$

2.2 Lensing Potential

The deflection angle derives from a 2D potential $\psi(\boldsymbol{\theta})$:

$$\boldsymbol{\alpha} = \nabla\psi, \quad \nabla^2\psi = 2\kappa \quad (6)$$

(C&K eqs. 4.14–4.15; N&B eq. 8). The 2D Poisson equation $\nabla^2\psi = 2\kappa$ is the central equation relating mass to deflection.

2.3 Jacobian Matrix and Magnification

The lens mapping $\boldsymbol{\theta} \rightarrow \boldsymbol{\beta}$ has the Jacobian matrix:

$$\mathbf{A} = \frac{\partial\boldsymbol{\beta}}{\partial\boldsymbol{\theta}} = \delta_{ij} - \frac{\partial^2\psi}{\partial\theta_i\partial\theta_j} = \begin{pmatrix} 1 - \kappa - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 - \kappa + \gamma_1 \end{pmatrix} \quad (7)$$

where γ_1 and γ_2 are the components of the **shear** $\gamma = \gamma_1 + i\gamma_2$ (C&K eq. 5.7; N&B eqs. 17–19).

The **magnification** is the inverse determinant:

Key Result

$$\mu = \frac{1}{\det \mathbf{A}} = \frac{1}{(1 - \kappa)^2 - |\gamma|^2} \quad (8)$$

2.4 Critical Curves and Caustics

Critical curves are the loci in the image plane where $\det \mathbf{A} = 0$ (infinite magnification). There are two types:

$$\text{Tangential: } 1 - \kappa - |\gamma| = 0 \quad (9)$$

$$\text{Radial: } 1 - \kappa + |\gamma| = 0 \quad (10)$$

The image of a critical curve under the lens mapping $\boldsymbol{\theta} \rightarrow \boldsymbol{\beta}$ is called a **caustic**. When a source crosses a caustic, images are created or destroyed in pairs (C&K Ch. 5; Petters et al. 2001).

3 Light Profiles: The Sérsic Function

Galaxy surface brightness is described by the **Sérsic profile** (Sérsic 1963):

$$I(R) = I_e \exp \left\{ -b_n \left[\left(\frac{R}{R_e} \right)^{1/n} - 1 \right] \right\} \quad (11)$$

where R_e is the effective (half-light) radius, n is the Sérsic index, and $b_n \approx 2n - 1/3 + 4/(405n)$ ensures that R_e encloses half the total light (Ciotti & Bertin 1999).

Special cases:

- $n = 1$: exponential profile (disk galaxies)
- $n = 4$: de Vaucouleurs profile (elliptical galaxies)

3.1 Ellipticity Parameterization

PyAutoLens parameterizes ellipticity using components (ϵ_1, ϵ_2) rather than axis ratio q and position angle ϕ :

$$\epsilon_1 = \frac{1-q}{1+q} \sin(2\phi), \quad \epsilon_2 = \frac{1-q}{1+q} \cos(2\phi) \quad (12)$$

with total ellipticity $|\epsilon| = (1-q)/(1+q)$. This avoids discontinuities at $\phi = 0, \pi$ and is standard in weak lensing (N&B Sec. 4.1).

4 Mass Profiles

4.1 Singular Isothermal Sphere (SIS)

The SIS has density $\rho(r) = \sigma_v^2/(2\pi G r^2)$, giving:

$$\kappa(\theta) = \frac{\theta_E}{2\theta} \quad (13)$$

$$\alpha(\theta) = \theta_E \quad (\text{constant!}) \quad (14)$$

with Einstein radius (C&K eq. 4.28):

$$\theta_E = 4\pi \left(\frac{\sigma_v}{c} \right)^2 \frac{D_{ds}}{D_s} \quad (15)$$

The mean convergence inside θ_E is $\langle \kappa \rangle = 1$, which is the defining property of the Einstein radius.

4.2 Singular Isothermal Ellipsoid (SIE)

The SIE generalizes the SIS to elliptical isodensity contours (Kormann, Schneider & Bartelmann 1994). It is the workhorse model for galaxy-scale lenses because it produces the correct image configurations (doubles and quads) with physically motivated mass distributions.

In PyAutoLens, the SIE is `al.mp.Isothermal` with parameters: `centre`, `ell_comps`, `einstein_radius`.

4.3 NFW Profile

The Navarro-Frenk-White (1996) profile describes dark matter halos:

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1+r/r_s)^2} \quad (16)$$

with known analytic convergence (Bartelmann 1996; C&K Sec. 4.5).

5 Multi-Plane Lensing

For mass at multiple redshifts, light is deflected sequentially at each lens plane. The multi-plane lens equation (S92, Ch. 9):

$$\boldsymbol{\theta}_{j+1} = \boldsymbol{\theta}_1 - \sum_{i=1}^j \beta_{ij} \boldsymbol{\alpha}_i(\boldsymbol{\theta}_i) \quad (17)$$

where $\beta_{ij} = D_{ij}D_s/(D_jD_{is})$ are the distance ratios. PyAutoLens handles this automatically via the `Tracer` object.

6 Cosmological Distances

All lensing distances are **angular diameter distances** in a Friedmann-Lemaître-Robertson-Walker cosmology:

$$D_A(z_1, z_2) = \frac{1}{1+z_2} \int_{z_1}^{z_2} \frac{c dz'}{H(z')} \quad (18)$$

for a flat universe, where $H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$.

PyAutoLens Implementation

PyAutoLens wraps **astropy** cosmology:

```
cosmo = al.cosmo.Planck15()
D_d = cosmo.angular_diameter_distance_to_earth_in_kpc_from(redshift
=0.5)
Sigma_cr = cosmo.
    critical_surface_density_between_redshifts_solar_mass_per_kpc2_from
    (
        redshift_0=0.5, redshift_1=1.0
    )
```

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